Supplementary Information: Validation controlled imbalance aware learning for fine grained IoT intrusion detection

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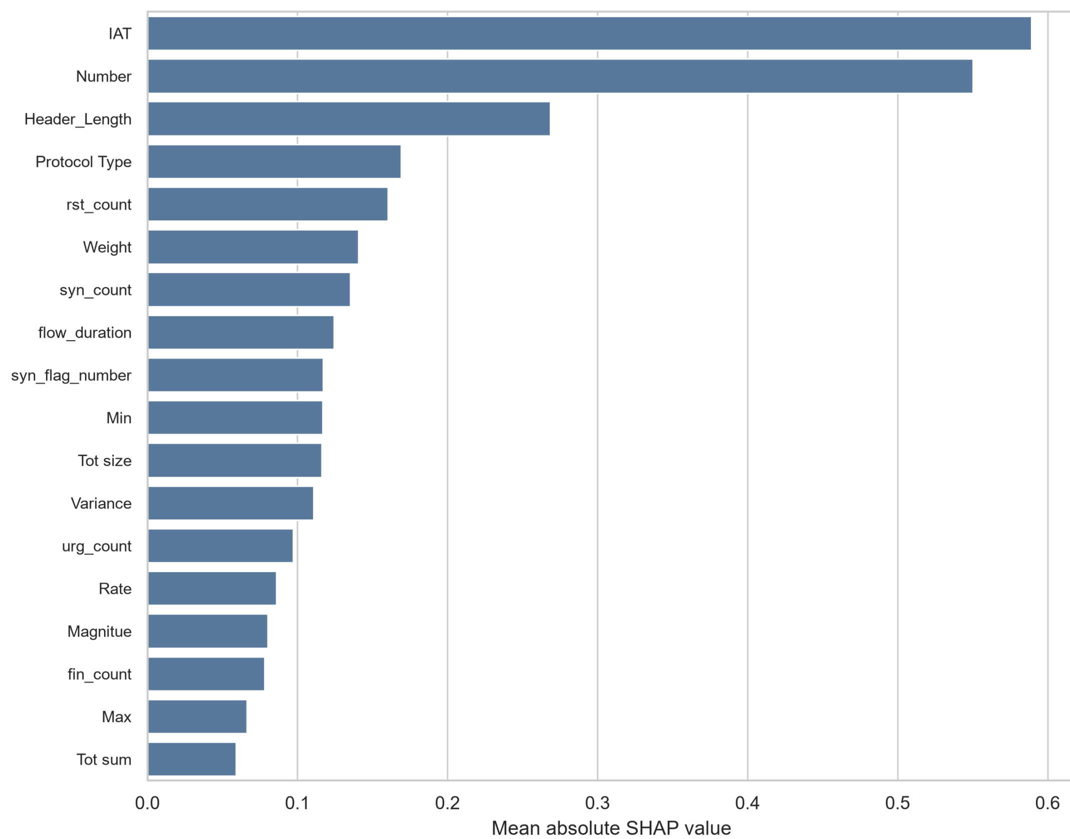
Supplementary Methods

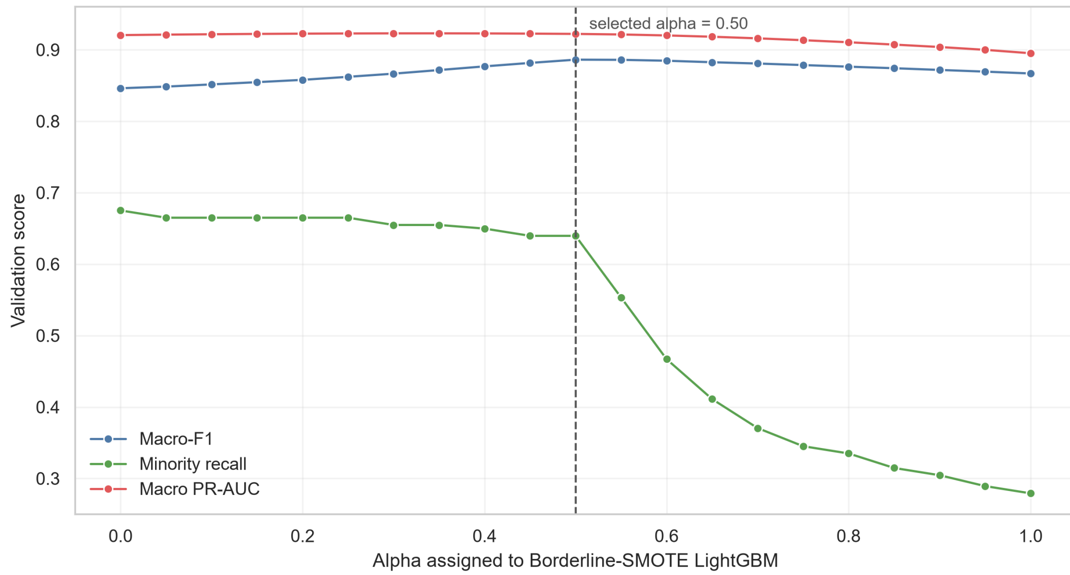
This supplementary file records overflow tables, result sources and reproduction commands for the Scientific Reports LaTeX submission package. Complete CSV files are supplied in the `supplementary_tables/` directory.

Supplementary Figure S1. CICIoT2023 SHAP feature contributions



Mean absolute SHAP values for the highest-ranked features in the CICIoT2023 Borderline-SMOTE LightGBM component. The analysis is used for component-level attribution, not causal interpretation.

Supplementary Figure S2. CICIoT2023 alpha sensitivity



Validation-subsplit Macro-F1, minority recall and macro PR-AUC across candidate ensemble weights. The selected value was $\alpha = 0.50$ for the Borderline-SMOTE LightGBM component.

Supplementary Table S1. CICIoT2023 baseline variants

Model	Accuracy	Balanced accuracy	Macro-F1	Minority recall	Macro PR-AUC
Flat LightGBM	0.9462	0.8446	0.8612	0.1910	0.8863
Random oversampling LightGBM	0.9465	0.8439	0.8602	0.2247	0.8841
Borderline-SMOTE LightGBM	0.9477	0.8495	0.8674	0.2584	0.8933
SMOTE LightGBM	0.9469	0.8452	0.8622	0.2097	0.8902
Weighted XGBoost	0.9359	0.8981	0.8453	0.6592	0.9203
Validation-controlled ensemble	0.9486	0.8883	0.8828	0.5131	0.9219

Supplementary Table S2. IoTID20 split-bootstrap deltas

Run ID	Metric	Mean delta	2.5% quantile	97.5% quantile	Splits	Resamples
iotid_fine_lgbm_mi_top10_weighted	balanced_accuracy	0.0568	0.0546	0.0592	10	10000
iotid_fine_lgbm_xgb_validation_ensemble	balanced_accuracy	0.0566	0.0542	0.0591	10	10000
iotid_fine_xgb_weighted	balanced_accuracy	0.0548	0.0527	0.0572	10	10000
iotid_fine_lgbm_weighted	balanced_accuracy	0.0533	0.0511	0.0556	10	10000
iotid_fine_lgbm_mi_top10_flat	balanced_accuracy	0.0033	0.0021	0.0045	10	10000
iotid_fine_lgbm_mi_top10_weighted	f1_macro	0.0140	0.0117	0.0165	10	10000
iotid_fine_lgbm_xgb_validation_ensemble	f1_macro	0.0138	0.0114	0.0163	10	10000
iotid_fine_lgbm_weighted	f1_macro	0.0123	0.0102	0.0144	10	10000
iotid_fine_xgb_weighted	f1_macro	0.0116	0.0094	0.0139	10	10000
iotid_fine_lgbm_mi_top10_flat	f1_macro	0.0031	0.0020	0.0042	10	10000
iotid_fine_lgbm_weighted	minority_recall_mean	0.2162	0.2117	0.2214	10	10000
iotid_fine_xgb_weighted	minority_recall_mean	0.2132	0.2099	0.2163	10	10000
iotid_fine_lgbm_mi_top10_weighted	minority_recall_mean	0.2129	0.2075	0.2178	10	10000
iotid_fine_lgbm_xgb_validation_ensemble	minority_recall_mean	0.2128	0.2077	0.2173	10	10000
iotid_fine_lgbm_mi_top10_flat	minority_recall_mean	-0.0012	-0.0017	-0.0008	10	10000
iotid_fine_lgbm_mi_top10_flat	pr_auc_macro	0.0019	0.0016	0.0022	10	10000
iotid_fine_lgbm_mi_top10_weighted	pr_auc_macro	0.0001	-0.0002	0.0004	10	10000
iotid_fine_lgbm_xgb_validation_ensemble	pr_auc_macro	0.0001	-0.0002	0.0004	10	10000
iotid_fine_lgbm_weighted	pr_auc_macro	-0.0016	-0.0018	-0.0014	10	10000
iotid_fine_xgb_weighted	pr_auc_macro	-0.0023	-0.0027	-0.0021	10	10000

Supplementary Table S3. IoTID20 exact sign-flip tests

Run ID	Metric	Mean delta	Positive splits	Negative splits	Zero splits	Exact two-sided p
iotid_fine_lgbm_mi_top10_weighted	f1_macro	0.0140	10	0	0	0.0020
iotid_fine_lgbm_xgb_validation_ensemble	f1_macro	0.0138	10	0	0	0.0020
iotid_fine_lgbm_weighted	f1_macro	0.0123	10	0	0	0.0020

Run ID	Metric	Mean delta	Positive splits	Negative splits	Zero splits	Exact two-sided p
iotid_fine_xgb_weighted	f1_macro	0.0116	10	0	0	0.0020
iotid_fine_lgbm_mi_top10_flat	f1_macro	0.0031	9	1	0	0.0039
iotid_fine_lgbm_weighted	minority_recall_mean	0.2162	10	0	0	0.0020
iotid_fine_xgb_weighted	minority_recall_mean	0.2132	10	0	0	0.0020
iotid_fine_lgbm_mi_top10_weighted	minority_recall_mean	0.2129	10	0	0	0.0020
iotid_fine_lgbm_xgb_validation_ensemble	minority_recall_mean	0.2128	10	0	0	0.0020
iotid_fine_lgbm_mi_top10_flat	minority_recall_mean	-0.0012	0	9	1	0.0039
iotid_fine_lgbm_mi_top10_flat	pr_auc_macro	0.0019	10	0	0	0.0020
iotid_fine_lgbm_mi_top10_weighted	pr_auc_macro	0.0001	5	5	0	0.5156
iotid_fine_lgbm_xgb_validation_ensemble	pr_auc_macro	0.0001	5	5	0	0.6523
iotid_fine_lgbm_weighted	pr_auc_macro	-0.0016	0	10	0	0.0020
iotid_fine_xgb_weighted	pr_auc_macro	-0.0023	0	10	0	0.0020

Supplementary Table S4. CICIoT2023 SHAP feature contributions

Rank	Feature	Mean absolute SHAP	LightGBM gain
1	IAT	0.5893	35188.0
2	Number	0.5501	9861.0
3	Header_Length	0.2686	27253.0
4	Protocol Type	0.1689	18648.0
5	rst_count	0.1605	18709.0
6	Weight	0.1408	3422.0
7	syn_count	0.1354	18901.0
8	flow_duration	0.1243	29979.0
9	syn_flag_number	0.1171	1138.0
10	Min	0.1168	17946.0
11	Tot size	0.1162	15534.0
12	Variance	0.1106	9005.0
13	urg_count	0.0971	21191.0
14	Rate	0.0859	31449.0
15	Magnitue	0.0803	8659.0
16	fin_count	0.0780	7324.0
17	Max	0.0662	11669.0
18	Tot sum	0.0590	11090.0
19	Covariance	0.0551	11545.0
20	Duration	0.0480	24389.0

Supplementary Table S5. CICIoT2023 bootstrap summary

Quantity	Point estimate	2.5% quantile	97.5% quantile	Resamples
baseline_macro_f1	0.8612	0.8580	0.8642	2000
baseline_balanced_accuracy	0.8446	0.8417	0.8477	2000
baseline_minority_recall	0.1910	0.1429	0.2385	2000
baseline_worst_class_recall	0.1910	0.1429	0.2385	2000
ensemble_macro_f1	0.8828	0.8801	0.8853	2000
ensemble_balanced_accuracy	0.8883	0.8854	0.8914	2000
ensemble_minority_recall	0.5131	0.4549	0.5714	2000
ensemble_worst_class_recall	0.5131	0.4549	0.5651	2000
delta_macro_f1	0.0216	0.0186	0.0246	2000
delta_balanced_accuracy	0.0437	0.0410	0.0465	2000
delta_minority_recall	0.3221	0.2635	0.3837	2000
delta_worst_class_recall	0.3221	0.2635	0.3798	2000

The full bootstrap sample table is supplied as `supplementary_tables/ciciot_bootstrap_samples.csv`.

Supplementary Table S6. CICIoT2023 alpha grid

Alpha	Macro-F1	Minority recall	Macro PR-AUC	Composite score
0.00	0.8462	0.6751	0.9206	0.9002
0.05	0.8485	0.6650	0.9212	0.9017
0.10	0.8515	0.6650	0.9218	0.9047
0.15	0.8547	0.6650	0.9222	0.9079
0.20	0.8579	0.6650	0.9226	0.9111
0.25	0.8622	0.6650	0.9228	0.9154
0.30	0.8666	0.6548	0.9230	0.9190
0.35	0.8717	0.6548	0.9230	0.9240
0.40	0.8768	0.6497	0.9229	0.9288
0.45	0.8816	0.6396	0.9226	0.9328
0.50	0.8862	0.6396	0.9222	0.9373
0.55	0.8859	0.5533	0.9214	0.9302
0.60	0.8847	0.4670	0.9201	0.9220
0.65	0.8826	0.4112	0.9183	0.9155
0.70	0.8808	0.3706	0.9160	0.9104
0.75	0.8786	0.3452	0.9134	0.9062
0.80	0.8764	0.3350	0.9106	0.9032
0.85	0.8742	0.3147	0.9073	0.8994
0.90	0.8718	0.3046	0.9039	0.8962
0.95	0.8695	0.2893	0.9000	0.8926
1.00	0.8668	0.2792	0.8952	0.8892

Supplementary Table S7. Runtime and model complexity

Dataset	Scope	Model	Train s	Inference ms per 10,000	Model size KB	Peak memory MB
CICIoT2023 public subsample	full confirmatory split	Flat LightGBM	136.1	1612.6	41698.0	not measured
CICIoT2023 public subsample	full confirmatory split	Borderline-SMOTE LightGBM	156.7	1715.0	41656.2	not measured
CICIoT2023 public subsample	full confirmatory split	Weighted XGBoost	278.0	275.6	24288.8	not measured
CICIoT2023 public subsample	full confirmatory split	Validation controlled ensemble	392.7	894.1	65945.0	not measured
IoTID20 public mirror	mean across 10 repeated 70/30 splits	Top ten weighted LightGBM	8.3	183.7	not aggregated	not measured
IoTID20 public mirror	mean across 10 repeated 70/30 splits	Validation controlled ensemble	26.4	125.5	not aggregated	not measured
IoTID20 public mirror	mean across 10 repeated 70/30 splits	Weighted LightGBM	12.2	189.9	not aggregated	not measured
IoTID20 public mirror	mean across 10 repeated 70/30 splits	Weighted XGBoost	18.6	65.8	not aggregated	not measured
IoTID20 public mirror	mean across 10 repeated 70/30 splits	Top ten flat LightGBM	8.1	183.3	not aggregated	not measured
IoTID20 public mirror	mean across 10 repeated 70/30 splits	Flat LightGBM	11.9	187.5	not aggregated	not measured
CICIoT2023 public subsample	stratified quick screen, train_n=220000, test_n=80000	Flat MLP	5.8	9.3	131.2	not measured
IoTID20 public mirror	stratified 70/30 quick screen	Flat MLP	11.7	6.9	56.8	not measured

Peak memory was not measured in the current reproducibility run. This is retained as a transparent limitation rather than replaced with an estimate.

Supplementary Table S8. Hyperparameters and implementation details

Core hyperparameters are supplied in `supplementary_tables/HYPERPARAMETER_TABLE.md`. The final CICIoT2023 public subsample ensemble used Borderline-SMOTE LightGBM and class weighted XGBoost probability outputs with validation selected alpha equal to 0.50. Borderline-SMOTE used adaptive `k_neighbors` from 1 to 5 according to the smallest class in the training split and the imbalanced learn default `m_neighbors`.

Supplementary Table S9. Short labels for Figure 2b

Display label	Original CICIoT2023 public subsample label
Uploading	Uploading_Attack
Ping sweep	Recon-PingSweep
Backdoor	Backdoor_Malware
XSS	XSS

Supplementary Table S10. Device codes for Figure 4

Device code	Original N-BaIoT held out device name
D1	Danmini_Doorbell
D2	Ecobee_Thermostat
D3	Ennio_Doorbell
D4	Philips_B120N10_Baby_Monitor
D5	Provision_PT_737E_Security_Camera
D6	Provision_PT_838_Security_Camera
D7	Samsung_SNH_1011_N_Webcam
D8	SimpleHome_XCS7_1002_WHT_Security_Camera
D9	SimpleHome_XCS7_1003_WHT_Security_Camera

Supplementary Table S11. Benchmark context against recent intrusion detection studies

Study	Dataset or task	Model family	Imbalance handling	Validation and boundary test	Uncertainty	Main limitation for direct comparison	Present distinction
Alsubaei 2025	NSL-KDD, UNSW-NB15, CICIDS2017 and related IDS benchmarks	XGBoost and optimized sequential neural network	Class imbalance considered through model and data choices	Cross validation and train test experiments; no split bootstrap or device held out test reported in the extracted manuscript	Not reported as paired uncertainty	Different datasets and task definitions; headline scores not directly comparable	Present study adds rare recall, frozen test selection, IoTID20 repeated holdouts and N-BaIoT device holdouts
Yu et al. 2025	NSL-KDD and CIC-IDS2017	Feature selection with CNN-LSTM	Equalized loss style class imbalance treatment	Repeated or fold based evaluation; no independent device holdout reported in the extracted manuscript	No paired split uncertainty table found	Strong architecture focus but less emphasis on frozen validation selection	Present study focuses on validation discipline and rare label diagnostics
Amine et al. 2025	IoT intrusion detection benchmarks	Deep learning based IDS	Benchmark specific imbalance handling	Extensive benchmark figures and tables	No row bootstrap or split bootstrap reported in the extracted manuscript	Model novelty focus; validation boundary less explicit	Present study treats validation controls as the main contribution
Ma et al. 2025	NF-ToN-IoT-v2, NF-UNSW-NB15-v2, NF-BoT-IoT-v2, NF-CSE-CIC-IDS2018-v2 and CIC-ToN-IoT	Feature selection, synthetic samples and LightGBM	Focal loss and synthetic minority support	Multiple datasets and train test validation	No paired uncertainty table found in the extracted manuscript	Strong multi dataset framing, but incompatible feature spaces and tasks	Present study complements this route with split and device boundary tests
Sarwar et al. 2025	N-BaIoT and UNSW-NB15	Decision tree, SVM, random forest and neural network	Feature selection and conventional classifiers	Five fold style evaluation and runtime reporting	No leave one device out N-BaIoT boundary reported in the extracted manuscript	Strong deployment context, but device transfer boundary is less direct	Present study uses complete held out devices in N-BaIoT
Singh et al. 2025	Smart grid intrusion detection data	Attentional LSTM ensemble	Severe minority class treatment with large recall gains in its task	Domain specific training and evaluation	Uncertainty treatment differs from this study	Different domain and label granularity, so scores are not a direct CICIoT2023 comparison	Present study positions rare recall gains under public IoT benchmark validation
Abu-Shareha et al. 2026	Supervised IDS literature	Review and multi criteria evaluation	Reviews imbalance as an evaluation dimension	Not an empirical detector	Not applicable	No new model result	Present study supplies an empirical validation controlled case

This matrix is a benchmark context rather than a leaderboard. The studies differ in datasets, label granularity, task definitions and split protocols, so headline accuracy and detection rate are not directly comparable with the CICIoT2023 public subsample Macro-F1 reported in the main manuscript.

Supplementary Table S12. Model family and baseline screen

Dataset	Scope	Model	Features	Macro-F1	Minority recall	Macro PR-AUC	Train s	Inference ms per 10,000
CICIoT2023 public subsample	stratified quick three component ensemble screen, 220000 train and 80000 test rows	Three component ensemble screen	46	0.8875	0.5733	0.9152	310.7	1559.2
CICIoT2023 public subsample	stratified quick three component ensemble screen, 220000 train and 80000 test rows	Three component ensemble screen	46	0.8875	0.5733	0.9152	310.7	1559.2
CICIoT2023 public subsample	stratified quick three component ensemble screen, 220000 train and 80000 test rows	Three component ensemble screen	46	0.8875	0.5733	0.9152	310.7	1559.2

Dataset	Scope	Model	Features	Macro-F1	Minority recall	Macro PR-AUC	Train s	Inference ms per 10,000
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Borderline-SMOTE LightGBM	46	0.8797	0.4464	0.9058	31.0	1393.0
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Weighted XGBoost	46	0.8637	0.5536	0.9127	45.4	239.5
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Hierarchical LightGBM	20	0.8565	0.5536	–	14.5	11758.0
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Flat LightGBM	46	0.8473	0.1250	0.8747	27.1	1345.0
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Weighted Random Forest	46	0.8421	0.5536	0.8702	8.1	166.1
CICIoT2023 public subsample	stratified quick SMOTEENN screen, 220000 train and 80000 test rows	SMOTEENN LightGBM	46	0.8082	0.7067	0.8495	86.8	1394.8
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Weighted CatBoost	46	0.7921	0.2857	0.8435	108.1	13.7
CICIoT2023 public subsample	stratified quick model family screen, 220000 train and 80000 test rows	Weighted ExtraTrees	46	0.7707	0.2321	0.8112	6.2	176.1
CICIoT2023 public subsample	stratified quick screen, train_n=220000, test_n=80000	Flat MLP	46	0.5361	0.0000	0.5744	5.8	9.3
IoTID20 public mirror	stratified 70/30 quick model family screen	Top ten weighted LightGBM	10	0.6209	0.4908	0.6784	9.9	204.0
IoTID20 public mirror	stratified 70/30 quick model family screen	Weighted LightGBM	22	0.6186	0.4986	0.6768	14.1	209.2
IoTID20 public mirror	stratified 70/30 quick model family screen	Weighted XGBoost	22	0.6177	0.4803	0.6764	21.5	73.5
IoTID20 public mirror	stratified 70/30 quick model family screen	Weighted CatBoost	22	0.6163	0.4504	0.6786	56.9	6.4
IoTID20 public mirror	stratified 70/30 quick model family screen	Flat LightGBM	22	0.6103	0.2637	0.6784	14.0	215.3
IoTID20 public mirror	stratified 70/30 quick model family screen	Weighted Random Forest	22	0.6054	0.4689	0.6562	15.0	69.7
IoTID20 public mirror	stratified 70/30 quick screen	Flat MLP	22	0.4937	0.0380	0.5750	11.7	6.9

The model family screen includes quick CICIoT2023 public subsample and IoTID20 checks, additional tree baselines and a simple MLP baseline. It is not used to replace the full frozen CICIoT2023 public subsample result in the main manuscript.

Supplementary Table S13. CICIoT2023 public subsample hardest and easiest class diagnostics

Diagnostic group	Class	Support	Flat recall	Ensemble recall	Recall delta	Flat F1	Ensemble F1
hardest by ensemble recall	Uploading_Attack	267	0.1910	0.5131	0.3221	0.2506	0.4521
hardest by ensemble recall	XSS	729	0.3827	0.5761	0.1934	0.4282	0.5357
hardest by ensemble recall	Backdoor_Malware	640	0.4719	0.6406	0.1688	0.5167	0.6129
hardest by ensemble recall	CommandInjection	1086	0.4401	0.6621	0.2219	0.5010	0.6815
hardest by ensemble recall	Recon-PingSweep	465	0.4065	0.6667	0.2602	0.5136	0.6072
hardest by ensemble recall	Recon-OSScan	9953	0.6749	0.6995	0.0246	0.7435	0.7592
hardest by ensemble recall	SqlInjection	1028	0.5409	0.7208	0.1800	0.6077	0.6767
hardest by ensemble recall	BrowserHijacking	1155	0.6667	0.7273	0.0606	0.7523	0.7116
hardest by ensemble recall	DictionaryBruteForce	2543	0.7157	0.7703	0.0547	0.7555	0.7283
hardest by ensemble recall	Recon-PortScan	9934	0.7725	0.7746	0.0021	0.7656	0.7772
hardest by ensemble recall	DNS_Spoofing	9990	0.7605	0.7855	0.0250	0.7882	0.8003
hardest by ensemble recall	MITM-ArpSpoofing	10072	0.8402	0.8417	0.0016	0.8819	0.8823
easiest non benign by ensemble recall	DDoS-RSTFINFlood	10162	1.0000	1.0000	0.0000	1.0000	1.0000
easiest non benign by ensemble recall	Mirai-udpplain	10074	0.9997	0.9999	0.0002	0.9999	0.9999
easiest non benign by ensemble recall	DDoS-PSHACK_Flood	10042	0.9999	0.9999	0.0000	1.0000	1.0000
easiest non benign by ensemble recall	DDoS-ICMP_Flood	9925	0.9999	0.9999	0.0000	0.9999	0.9999
easiest non benign by ensemble recall	DDoS-SynonymousIP_Flood	9845	0.9999	0.9999	0.0000	0.9995	0.9995
easiest non benign by ensemble recall	DDoS-ACK_Fragmentation	9998	0.9998	0.9998	0.0000	0.9998	0.9998
easiest non benign by ensemble recall	DoS-TCP_Flood	10067	0.9997	0.9997	0.0000	0.9997	0.9997
easiest non benign by ensemble recall	DDoS-ICMP_Fragmentation	10088	0.9995	0.9996	0.0001	0.9998	0.9998
easiest non benign by ensemble recall	DDoS-UDP_Flood	10050	0.9996	0.9996	0.0000	0.9995	0.9996
easiest non benign by ensemble recall	DDoS-TCP_Flood	9967	0.9997	0.9996	-0.0001	0.9997	0.9997

Diagnostic group	Class	Support	Flat recall	Ensemble recall	Recall delta	Flat F1	Ensemble F1
easiest non benign by ensemble recall	Mirai-greip_flood	10076	0.9995	0.9995	0.0000	0.9990	0.9993
easiest non benign by ensemble recall	DoS-UDP_Flood	9982	0.9993	0.9995	0.0002	0.9995	0.9996

Supplementary Table S14. CICIoT2023 public subsample low support class diagnostics

Class	Support	Flat precision	Flat recall	Flat F1	Ensemble precision	Ensemble recall	Ensemble F1
Uploading_Attack	267	0.3643	0.1910	0.2506	0.4041	0.5131	0.4521
Recon-PingSweep	465	0.6974	0.4065	0.5136	0.5576	0.6667	0.6072
Backdoor_Malware	640	0.5709	0.4719	0.5167	0.5874	0.6406	0.6129
XSS	729	0.4861	0.3827	0.4282	0.5006	0.5761	0.5357
SqlInjection	1028	0.6933	0.5409	0.6077	0.6377	0.7208	0.6767
CommandInjection	1086	0.5815	0.4401	0.5010	0.7021	0.6621	0.6815
BrowserHijacking	1155	0.8632	0.6667	0.7523	0.6965	0.7273	0.7116
DictionaryBruteForce	2543	0.8000	0.7157	0.7555	0.6905	0.7703	0.7283
DDoS-SlowLoris	4642	0.9966	0.9974	0.9970	0.9970	0.9991	0.9981
DDoS-HTTP_Flood	5766	0.9988	0.9964	0.9976	0.9986	0.9967	0.9977
VulnerabilityScan	7546	0.9984	0.9991	0.9987	0.9985	0.9991	0.9988
DDoS-SynonymousIP_Flood	9845	0.9991	0.9999	0.9995	0.9992	0.9999	0.9995

Supplementary Table S15. IoTID20 repeated split class wise diagnostics

Class	Model	Mean support	Mean recall	SD recall	Mean F1	Recall delta	F1 delta
DoS-Synflooding	Flat LightGBM	17817.0	0.9992	0.0002	0.9995	-0.0001	-0.0001
DoS-Synflooding	Top ten flat LightGBM	17817.0	0.9992	0.0002	0.9995	-0.0001	-0.0001
DoS-Synflooding	Top ten weighted LightGBM	17817.0	0.9992	0.0002	0.9995	-0.0001	-0.0001
DoS-Synflooding	Validation controlled ensemble	17817.0	0.9992	0.0002	0.9995	-0.0001	-0.0001
DoS-Synflooding	Weighted XGBoost	17817.0	0.9992	0.0002	0.9995	-0.0001	-0.0001
MITM ARP Spoofing	Flat LightGBM	10613.0	0.4659	0.0042	0.5535	0.2042	0.0045
MITM ARP Spoofing	Top ten flat LightGBM	10613.0	0.4612	0.0032	0.5504	0.2042	0.0045
MITM ARP Spoofing	Top ten weighted LightGBM	10613.0	0.6701	0.0047	0.5580	0.2042	0.0045
MITM ARP Spoofing	Validation controlled ensemble	10613.0	0.6706	0.0043	0.5578	0.2042	0.0045
MITM ARP Spoofing	Weighted XGBoost	10613.0	0.6731	0.0045	0.5446	0.2042	0.0045
Mirai-Ackflooding	Flat LightGBM	16537.0	0.2161	0.0386	0.2251	0.3510	0.1543
Mirai-Ackflooding	Top ten flat LightGBM	16537.0	0.2333	0.0353	0.2437	0.3510	0.1543
Mirai-Ackflooding	Top ten weighted LightGBM	16537.0	0.5671	0.0470	0.3794	0.3510	0.1543
Mirai-Ackflooding	Validation controlled ensemble	16537.0	0.5672	0.0480	0.3791	0.3510	0.1543
Mirai-Ackflooding	Weighted XGBoost	16537.0	0.5340	0.0313	0.3720	0.3510	0.1543
Mirai-HTTP Flooding	Flat LightGBM	16746.0	0.2463	0.0431	0.2497	0.0203	0.0152
Mirai-HTTP Flooding	Top ten flat LightGBM	16746.0	0.2544	0.0478	0.2609	0.0203	0.0152
Mirai-HTTP Flooding	Top ten weighted LightGBM	16746.0	0.2666	0.0453	0.2648	0.0203	0.0152
Mirai-HTTP Flooding	Validation controlled ensemble	16746.0	0.2644	0.0449	0.2631	0.0203	0.0152
Mirai-HTTP Flooding	Weighted XGBoost	16746.0	0.2977	0.0270	0.2829	0.0203	0.0152
Mirai-Hostbruteforceg	Flat LightGBM	36354.0	0.7673	0.0047	0.7084	-0.2632	-0.0559
Mirai-Hostbruteforceg	Top ten flat LightGBM	36354.0	0.7618	0.0052	0.7049	-0.2632	-0.0559
Mirai-Hostbruteforceg	Top ten weighted LightGBM	36354.0	0.5041	0.0037	0.6526	-0.2632	-0.0559
Mirai-Hostbruteforceg	Validation controlled ensemble	36354.0	0.5038	0.0039	0.6524	-0.2632	-0.0559
Mirai-Hostbruteforceg	Weighted XGBoost	36354.0	0.4861	0.0029	0.6379	-0.2632	-0.0559
Mirai-UDP Flooding	Flat LightGBM	55066.0	0.7681	0.0134	0.7761	-0.0644	0.0500
Mirai-UDP Flooding	Top ten flat LightGBM	55066.0	0.7760	0.0157	0.7816	-0.0644	0.0500
Mirai-UDP Flooding	Top ten weighted LightGBM	55066.0	0.7037	0.0012	0.8261	-0.0644	0.0500
Mirai-UDP Flooding	Validation controlled ensemble	55066.0	0.7037	0.0012	0.8261	-0.0644	0.0500
Mirai-UDP Flooding	Weighted XGBoost	55066.0	0.7037	0.0012	0.8261	-0.0644	0.0500
Normal	Flat LightGBM	12022.0	0.9697	0.0010	0.9840	0.0008	-0.0007
Normal	Top ten flat LightGBM	12022.0	0.9698	0.0009	0.9839	0.0008	-0.0007
Normal	Top ten weighted LightGBM	12022.0	0.9705	0.0009	0.9833	0.0008	-0.0007
Normal	Validation controlled ensemble	12022.0	0.9705	0.0009	0.9833	0.0008	-0.0007
Normal	Weighted XGBoost	12022.0	0.9687	0.0008	0.9823	0.0008	-0.0007
Scan Hostport	Flat LightGBM	6658.0	0.2691	0.0030	0.4011	0.2129	-0.0416
Scan Hostport	Top ten flat LightGBM	6658.0	0.2679	0.0030	0.3998	0.2129	-0.0416
Scan Hostport	Top ten weighted LightGBM	6658.0	0.4820	0.0081	0.3595	0.2129	-0.0416
Scan Hostport	Validation controlled ensemble	6658.0	0.4819	0.0078	0.3596	0.2129	-0.0416
Scan Hostport	Weighted XGBoost	6658.0	0.4824	0.0065	0.3583	0.2129	-0.0416
Scan Port OS	Flat LightGBM	15922.0	0.6736	0.0139	0.5760	0.0492	0.0007
Scan Port OS	Top ten flat LightGBM	15922.0	0.6808	0.0098	0.5768	0.0492	0.0007
Scan Port OS	Top ten weighted LightGBM	15922.0	0.7228	0.0066	0.5766	0.0492	0.0007

Class	Model	Mean support	Mean recall	SD recall	Mean F1	Recall delta	F1 delta
Scan Port OS	Validation controlled ensemble	15922.0	0.7231	0.0062	0.5767	0.0492	0.0007
Scan Port OS	Weighted XGBoost	15922.0	0.7236	0.0065	0.5744	0.0492	0.0007

Supplementary Table S16. N-BaIoT cap sensitivity

Cap	Task	Run ID	Folds	Mean F1	Min F1	Mean present F1	Min present recall	Mean PR-AUC
5000	attack	nb_attack_lgbm_importance_top40_weighted	9	0.9193	0.8137	0.9348	0.0014	0.9655
5000	attack	nb_attack_lgbm_mi_top40_weighted	9	0.9325	0.8143	0.9325	0.0014	0.9642
5000	attack	nb_attack_lgbm_weighted	9	0.9388	0.8526	0.9388	0.0014	0.9652
20000	attack	nb_attack_lgbm_weighted	9	0.9435	0.7483	0.9867	0.6917	0.9902
50000	attack	nb_attack_lgbm_weighted	9	0.9948	0.9855	0.9948	0.8767	0.9960
5000	binary	nb_binary_lgbm_weighted	9	0.9832	0.8613	0.9832	0.9355	–
20000	binary	nb_binary_lgbm_weighted	9	0.9998	0.9993	0.9998	0.9990	–
5000	family	nb_family_lgbm_weighted	9	0.9998	0.9992	0.9998	0.9984	1.0000
20000	family	nb_family_lgbm_weighted	9	0.9999	0.9995	0.9999	0.9990	1.0000

The 50,000 row per file sensitivity was run for the fine grained attack task because binary and family settings were already near saturated at the smaller caps.

Supplementary Table S17. CICIoT2023 rare prior operating point screen

Objective	Alpha	Rare scope	Multiplier	Accuracy	Macro-F1	Minority recall	Macro PR-AUC	Worst class
macro_f1	0.50	lowest_support	1.00	0.9450	0.8875	0.5733	0.9152	Recon-PingSweep
macro_f1_plus_minority	0.50	lowest_support	4.00	0.9447	0.8803	0.6000	0.9149	XSS
recall_sensitive	0.50	lowest_support	6.00	0.9446	0.8787	0.6400	0.9147	XSS
minority_recall_constrained	0.50	lowest_support	6.00	0.9446	0.8787	0.6400	0.9147	XSS

The source for this table is the quick stratified screen. Rare prior tuning multiplies probabilities for the lowest support label or labels selected from the training split and selects the multiplier on an internal validation split. It is reported as a recall sensitive operating point, not as a replacement for the primary Macro-F1 operating point.

Reproduction commands

```
.\.venv\Scripts\python.exe scripts\run_ciciot_confirmatory_experiments.py `
  --out-dir ciciot_confirmatory_full
.\.venv\Scripts\python.exe scripts\run_ciciot_recall_aware_ensemble.py `
  --out-dir ciciot_ensemble_full
.\.venv\Scripts\python.exe scripts\run_ciciot_rare_prior_tuning.py `
  --out-dir ciciot_rare_prior_tuning_full
.\.venv\Scripts\python.exe scripts\run_ciciot_smoteenn_ensemble.py `
  --train-n 220000 --test-n 80000 --out-dir ciciot_ensemble_smoteenn_quick
.\.venv\Scripts\python.exe scripts\run_iotid20_repeated_split_uncertainty.py
.\.venv\Scripts\python.exe scripts\run_nbaiot_unseen_device.py `
  --n-per-file 20000 --out-dir nbaiot_unseen_device_20k `
  --run-id nb_binary_lgbm_weighted `
  --run-id nb_family_lgbm_weighted `
  --run-id nb_attack_lgbm_weighted
.\.venv\Scripts\python.exe scripts\run_ciciot_explainability.py `
  --shap-sample 3000
.\.venv\Scripts\python.exe scripts\run_ciciot_bootstrap_uncertainty.py `
  --n-bootstrap 2000 --out-dir statistical_uncertainty
.\.venv\Scripts\python.exe scripts\run_mlp_baselines.py `
  --out-dir mlp_baselines
.\.venv\Scripts\python.exe scripts\make_result_figures.py
```